Zhengyang (Jason) Qi

Founding Research Scientist, Listen Labs

San Francisco Bay Area | jasonqi146@gmail.com | 404-797-9976

Website | Google Scholar | GitHub | LinkedIn

Research Interests

My recent research explores how people reason, make decisions, and form expectations about the future. In particular, I study how mid-training and reinforcement learning can help language models simulate these processes. I focus on scaling these simulations across diverse individuals and situations, and on building realistic human simulators for training and evaluating long-horizon agents.

Ultimately, I hope to build models that reflect the richness and diversity of human behavior, with the ambition of making realistic simulation possible for any person, in any context.

Education

Carnegie Mellon University, School of Computer Science Pittsburgh, PA

M.S. in Intelligent Information Systems, Language Technologies Institute Aug 2022 - May 2024

- Research: RL for language agents, social intelligence, and multimodal ML; advisor: Prof. Graham Neubig.

Georgia Institute of Technology and Emory University, Dual Degree Program Atlanta, GA

B.S. in Computer Science; B.A. in Mathematics and Economics Aug 2017 - May 2022

Publications

** denotes equal contribution.*

1. Yining Zhao, Bushi Liu, Haofei Yu, **Zhengyang Qi**, Shanyong Wang, Chuyue Li, Yuxiang Liu, Jiaxuan You. **Learning to Simulate Individuals from Macro Social Signals**. *arXiv preprint arXiv:2610.07062, 2026*. [Paper]
   Introduces macro2mind, which uses GRPO with prediction-market signals to learn behavioral reasoning that transfers to individual simulation.

2. **Zhengyang Qi**, Charles Andrew Dickens, Derek Pham, Amanda Dsouza, Armin Parchami, Frederic Sala, Paroma Varma. **RIFT: A Rubric Failure Mode Taxonomy and Automated Diagnostics**. *ICLR 2026 Data Problems for Foundation Models Workshop, 2026*. [Paper]
   Introduces a taxonomy and automated diagnostics for detecting rubric failure modes before they corrupt evaluation or reward signals.

3. Mengqi Yuan, Zilong Zhou, Xinzhuang Xiong, et al., **Zhengyang Qi** (Advisor), Vincent Sunn Chen (Advisor), Frederic Sala (Advisor), ..., Tao Yu. **OSWorld 2.0: Benchmarking Computer Use Agents on Long-Horizon Real-World Tasks**. *arXiv preprint, 2026*. [Paper]
   A benchmark of long-horizon, real-world computer-use tasks with verifiable end-state evaluation; advised project design, research direction, and paper writing.

4. Amanda Dsouza, Harit Vishwakarma, **Zhengyang Qi**, Justin Bauer, Derek Pham, Thomas Walshe, Armin Parchami, Frederic Sala, Paroma Varma. **Automating Benchmark Design**. *arXiv preprint arXiv:2510.25039, 2025*. [Paper]
   Presents BeTaL, an interactive LLM-in-the-loop framework for automatically generating model-specific benchmarks across difficulty levels.

5. Haofei Yu*, **Zhengyang Qi***, Yining Zhao*, Kolby Nottingham, Keyang Xuan, Bodhisattwa Prasad Majumder, Hao Zhu, Paul Pu Liang, Jiaxuan You. **Sotopia-RL: Reward Design for Social Intelligence**. *arXiv preprint arXiv:2508.03905, 2025*. [Paper]
   Studies reward design, multi-signal rewards, and long-horizon credit assignment for RL training of socially intelligent language agents.

6. Haofei Yu*, **Zhengyang Qi***, Lawrence Keunho Jang*, Ruslan Salakhutdinov, Louis-Philippe Morency, Paul Pu Liang. **MMoE: Enhancing Multimodal Models with Mixtures of Multimodal Interaction Experts**. *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, 2024*. [Paper]
   Trains specialized experts for distinct cross-modal interaction patterns to improve multimodal model performance.

7. Ruiyi Wang*, Haofei Yu*, Wenxin Zhang*, **Zhengyang Qi***, Maarten Sap, Yonatan Bisk, Graham Neubig, Hao Zhu. **SOTOPIA-π: Interactive Learning of Socially Intelligent Language Agents**. *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics, 2024*. [Paper]
   Improves social intelligence through interactive self-play and self-reinforcement learning.

8. Xuhui Zhou*, Hao Zhu*, Leena Mathur, Ruohong Zhang, Haofei Yu, **Zhengyang Qi**, Louis-Philippe Morency, Yonatan Bisk, Daniel Fried, Graham Neubig, Maarten Sap. **SOTOPIA: Interactive Evaluation for Social Intelligence in Language Agents**. *International Conference on Learning Representations, 2024. Spotlight*. [Paper]
   Introduces interactive environments and rubric-based evaluation for social intelligence in language agents.

9. Simon Stepputtis, Joseph Campbell, Yaqi Xie, **Zhengyang Qi**, Wenxin Zhang, Ruiyi Wang, Sanketh Rangreji, Charles Michael Lewis, Katia P. Sycara. **Long-Horizon Dialogue Understanding for Role Identification in the Game of Avalon with Large Language Models**. *Findings of the Association for Computational Linguistics: EMNLP, 2023*. [Paper]
   Evaluates hidden-role and deception reasoning in Avalon, showing that long-horizon social reasoning remains difficult for LLMs.

10. Hayeong Song, **Zhengyang Qi**, John Stasko, Diyi Yang. **Understanding People's Needs in Viewing Diverse Social Opinions about Controversial Topics**. *IEEE 16th Pacific Visualization Symposium, 2023*. [Paper]
    A user study of how people seek out and navigate diverse opinions on controversial topics, with implications for interface design.

Talks, Presentations, and Service

- **ACL Rolling Review (ARR).** Regular Reviewer. Ongoing service.
- **Conference Reviewing.** Reviewer for ICLR 2027, AAAI 2027, ICML 2026, NeurIPS 2026, ICLR 2026, AAAI 2026, and NeurIPS 2025.
- **Cal Hacks 2026.** Judge. Upcoming.
- **Snorkel AI Reading Group.** Host. Coordinated invited talks, introduced speakers, and co-led Q&A sessions; speakers included researchers from Stanford, UC Berkeley, and Google DeepMind. 2026 (series). [Post 1] [Post 2] [Post 3] [Event]
- **Daytona AI Researchers Meetup at Stanford.** Speaker. Stanford University; hosted with FounderCoHo. Talk on RL agents, post-training, evaluation, and AI infrastructure. Apr 29, 2026. [Post] [Agenda]
- **OpenEnv Hackathon.** Final Judge. Sponsored by Snorkel AI with Cerebral Valley and PyTorch; judged projects on environment design, model training, multi-agent reasoning, long-horizon planning, world modeling, and self-improvement. Mar 7-8, 2026. [Post] [Event]
- **Daytona x Sentry AI Builders Meetup.** Speaker. San Francisco; hosted in partnership with Sentry. Dec 9, 2025. [Post] [Agenda]
- **AGI House Self-Evolving Agents Build Day.** Speaker. Palo Alto; hands-on event for prototyping adaptive and self-improving AI systems. Nov 1, 2025. [Post] [Recap]

Industry Research Experience

Listen Labs San Francisco Bay Area

Founding Research Scientist Sep 2026 - Present

- Build longitudinal representations of human context that capture how people's experiences, preferences, and expectations evolve over time and shape their decisions.
- Develop reinforcement learning methods for post-training language models to simulate human reasoning and decision-making across diverse individuals and situations.
- Scale human simulation models to support realistic behavioral modeling and the training and evaluation of long-horizon agents.

Snorkel AI San Francisco, CA

Senior Research Scientist, Agent RL & Evaluation Research Sep 2025 - Sep 2026

- Designed and ran task curation and verification pipelines for SWE-Bench Pro, Terminal-Bench 2.0, OSWorld 2.0, PostTrainBench, and EDA computer-use agent tasks, spanning synthetic environment generation, verifier-validity checks, and LLM-as-judge and rubric-based review; delivered agent-task datasets, frontier coding-agent preference evaluations, and clinical-agent rubrics.
- Researched reward hacking, verifier validity, and rubric failure modes in agent environments and evaluation pipelines to improve data quality; built RIFT (ICLR 2026 workshop), a rubric failure-mode taxonomy and automated diagnostics for auditing evaluation and reward signals.
- Independently built the company's agentic RL training stack on verl and Kubernetes: asynchronous, fault-tolerant rollout workers, sandboxed terminal-environment execution, and multi-node training on 64 H200 GPUs with Ray, Megatron, and SGLang.
- Trained long-horizon terminal agents with GRPO, improving Terminal-Bench 2.0 from 5.6% to 13.4% (Qwen3-32B) and from 36% to 41% (Qwen3.5-27B) through environment selection, reward attribution with partial-credit assignment, rejection sampling, and on-policy distillation.
- Published on automated benchmark design and milestone-based evaluation for RL; advised research direction and environment/task design for OSWorld 2.0.

Scale AI San Francisco, CA

Machine Learning Research Engineer Aug 2024 - Aug 2025

- Led post-training of multimodal LLM judge and reward models with reasoning RL and RLHF (SFT, GRPO, and human preference supervision), curating 60k+ evaluation and 300k+ training examples across multilingual instruction following, multimodal reasoning, STEM, and coding; the production judge outperformed GPT-4o on human reward and preference alignment.
- Built an agentic auto-evaluation system for scalable oversight of human-AI data-generation pipelines; deployed on 10+ production pipelines, cutting human labeling cost by up to 60% while preserving calibration and human-judgment quality.
- Built automated quality-control agents for RLHF rubric pipelines (RAG, tool use, multi-agent debate), outperforming baseline agents across deployments.

Academic Research Experience

Carnegie Mellon University, Language Technologies Institute Pittsburgh, PA

Graduate Research Assistant; advisor: Prof. Graham Neubig Aug 2022 - May 2024

- Researched interactive evaluation, social intelligence, multimodal ML, and reinforcement learning for language agents; co-built the SOTOPIA interactive evaluation platform (ICLR 2024 Spotlight) and the SOTOPIA-π self-play and self-reinforcement training pipeline, leading to publications at ICLR, ACL, and EMNLP.
- Built research systems and datasets for simulated social environments, rubric-based evaluation, self-play training, multimodal interaction modeling (MMoE), and long-horizon dialogue understanding (Avalon).

Georgia Institute of Technology and Emory University Atlanta, GA

Undergraduate Researcher; advisors: Prof. Diyi Yang and Prof. Xu Chu Aug 2017 - May 2022

- Conducted research in social computing, NLP, and data-centric ML; published a user study of how Reddit users navigate controversial discussions, with design implications for diverse-opinion interfaces (IEEE PacificVis 2023).

Additional Industry Experience

Intuit

Software Engineer Intern, Machine Learning May 2023 - Aug 2023

- Built LLM and ML services for TurboTax tax-form de-duplication across 1k+ form variations, supporting large-scale filing workflows for 42M users.

Intuit

Software Engineer Intern, Machine Learning Jun 2022 - Aug 2022

- Built multimodal image-captioning and large-batch inference pipelines for corporate media management, including Spark data processing and CUDA/PyTorch optimization.

Kiwi

Co-Founder and Machine Learning Engineer Dec 2020 - Dec 2021

- Led development of a fashion recommendation and entity-matching system using text, image, and metadata from 18 retailer websites.

EPI-USE America, Inc. Atlanta, GA

Innovation Engineer Intern Jun 2021 - Aug 2021

- Developed a graphical interface for interactive visualization of information networks.
- Built a BERT-based privacy-leakage classifier that replaced 8+ hours of manual work per week.

TechX Academy Shanghai, China

Assistant Virtual Reality Instructor Jul 2017 - Aug 2017

Awards and Honors

- **ICML 2026 Silver Reviewer Award**. International Conference on Machine Learning, 2026.
- **CMU RISE Scholarship** ($5k research award). Carnegie Mellon University, 2023.
- **Top 1.6% Placement, Amex Default Prediction**. Kaggle; ensemble modeling and feature engineering, 2022.

Open-Source and Research Artifacts

- **Sotopia:** Core contributor to an open-ended social learning and evaluation environment for language agents, underlying the SOTOPIA, SOTOPIA-π, and Sotopia-RL lines of research.
- **Chinese FLAN-T5 / COIG:** Contributed 20+ data pipelines for instruction-following LLM training in Chinese and multilingual settings.

Technical Skills

- **Post-Training Data and Evaluation:** RLHF/RLAIF, GRPO, PPO, DPO, SFT, reward modeling, process rewards, LLM-as-judge, rubric evaluation, scalable oversight, synthetic data generation, benchmark and environment curation, reasoning RL, multimodal models.
- **ML and Systems:** PyTorch, Hugging Face, Ray, verl, vLLM, SGLang, CUDA, Docker, AWS, Kubernetes, Spark, FSDP, Megatron, DeepSpeed.
- **Programming Languages:** Python, Unix shell, C, C++, SQL, JavaScript, Java.